

Chapter 6 - Set Theory

Part 2: Algebraic Proofs and Boolean Algebra

Sections 6.3 - 6.4

COMP 233 Discrete Mathematics | Birzeit University

Where we are: Chapter 6 at a glance

6.1

Set Basics

Sets, subsets, operations, partitions, power sets, products

6.2

Properties of Sets

Set identities and the element method of proof

6.3

Algebraic Proofs

Disproof by counterexample; deriving identities algebraically

6.4

Boolean Algebra

The shared structure behind logic and sets

This deck covers 6.3 and 6.4 (6.1 and 6.2 were Part 1).

6.3 Disproofs and Algebraic Proofs - outline

- 1. Disproving a false set property by counterexample
- 2. The number of subsets: $|P(X)| = 2^n$
- 3. Algebraic proofs of set identities
- 4. Symmetric difference (enrichment)

Disproving a proposed set property

Principle

A property “for all sets A, B, C, ...” is a universal statement.

To DISPROVE it, exhibit ONE counterexample: specific sets that make the two sides unequal.

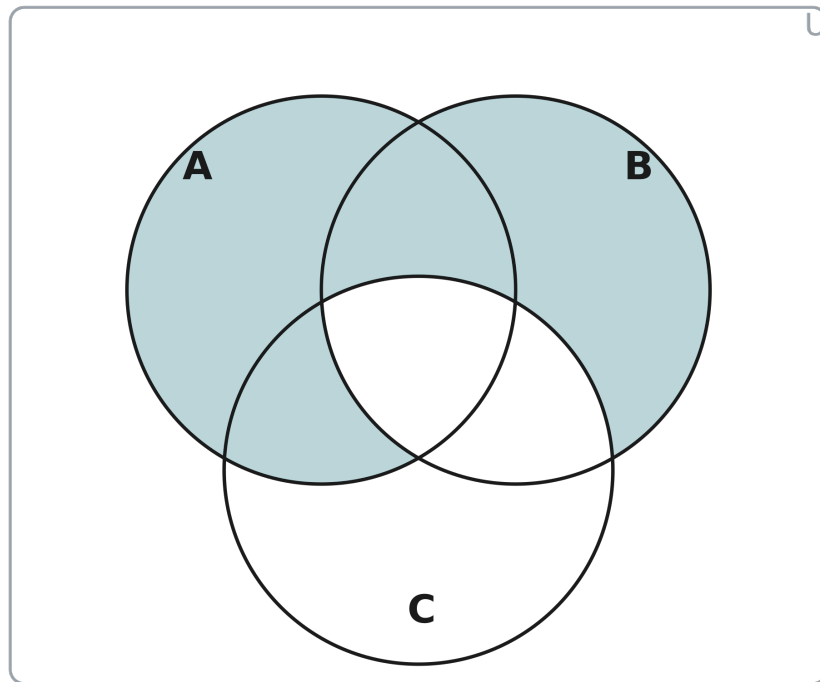
The claim we will test (Epp 6.3.1)

Is this true for all sets A, B, C ?

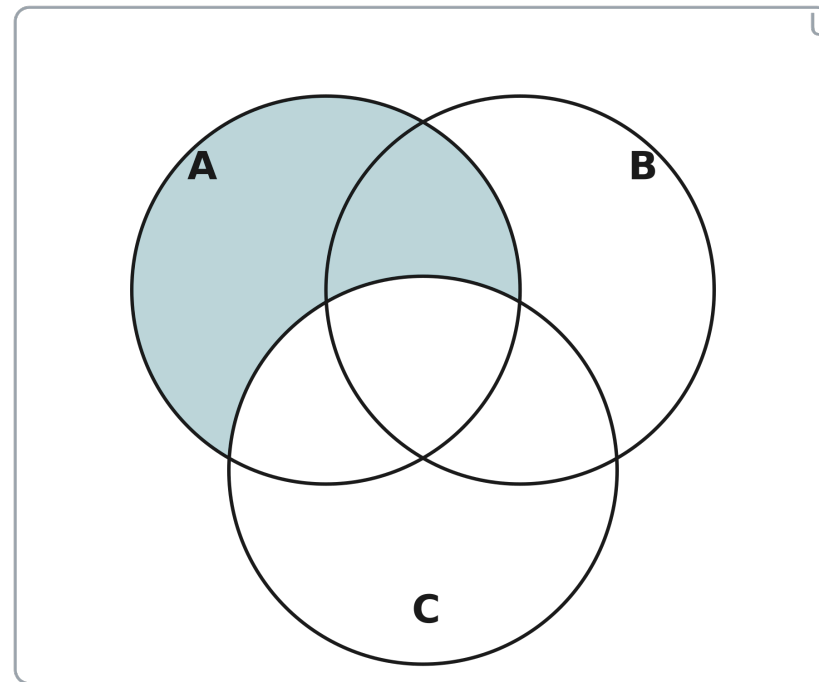
$$(A - B) \cup (B - C) = A - C$$

Strategy: shade both sides on a Venn diagram; if the regions differ, hunt for a counterexample.

Step 1: the two sides shade different regions



$$(A - B) \cup (B - C)$$



$$A - C$$

Different!

The left shades part of B that lies outside A and C; the right never shades anything outside A.

Step 2: a concrete counterexample settles it

Counterexample

Let $A = \emptyset$, $B = \{3\}$, $C = \emptyset$.

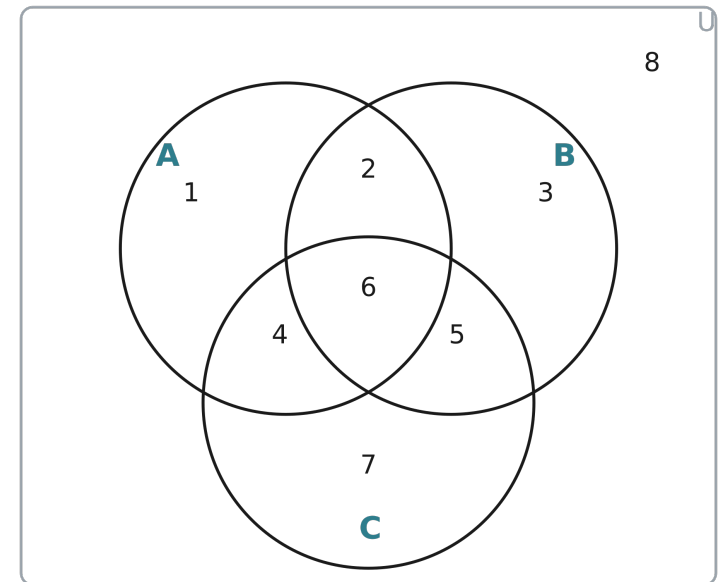
$$A - B = \emptyset \quad B - C = \{3\}$$

$$A - C = \emptyset$$

$$\text{Left: } (A - B) \cup (B - C) = \emptyset \cup \{3\} = \{3\}$$

$$\text{Right: } A - C = \emptyset$$

$$\{3\} \neq \emptyset$$



8 independent regions $\rightarrow$ Venn alone is informal

How many subsets does a set have?

Theorem 6.3.1

For every integer $n \geq 0$: if a set X has n elements, then $P(X)$ has 2^n elements.

n	example set	subsets	count = 2^n
0	$\emptyset$	$\emptyset$	1
1	$\{a\}$	$\emptyset, \{a\}$	2
2	$\{a, b\}$	$\emptyset, \{a\}, \{b\}, \{a, b\}$	4
3	$\{a, b, c\}$	... (all of them)	8

Each element is independently in or out of a subset: 2 choices, n times, gives 2^n .

Proof of $|P(X)| = 2^n$ by induction

Induction on n

Base $n = 0$:

$\emptyset$ has exactly one subset ($\emptyset$),

and $2^0 = 1$. ✓

Inductive step:

assume any k -element set has 2^k subsets.

Let X have $k+1$ elements; pick $z \in X$.

Subsets of X either omit z or contain z .

$\# \text{omit } z = \# \text{subsets of } X - \{z\} = 2^k$.

$A \mapsto A \cup \{z\}$ pairs them, so $\# \text{contain } z = 2^k$.

Total $= 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$. ✓

Pairing for $X = \{x, y, z\}$

omit z	contain z
$\emptyset$	$\{z\}$
$\{x\}$	$\{x, z\}$
$\{y\}$	$\{y, z\}$
$\{x, y\}$	$\{x, y, z\}$

$4 \leftrightarrow 4$, so $8 = 2 \cdot 4$ total.

Algebraic proofs: chaining known identities

The idea

Instead of arguing element by element, transform one side into the other using the identities of Theorem 6.2.2, citing one law per step.

It is like simplifying an algebraic expression in ordinary algebra - but the “numbers” are sets and the “operations” are $\cup$, $\cap$, and complement.

Most useful laws to keep ready

Distributive De Morgan Double complement

Set difference: $A - B = A \cap B^c$ *(this turns every “-” into $\cap$ with a complement)*

Worked algebraic proof 1

Prove for all sets A, B, C: $(A \cup B) - C = (A - C) \cup (B - C)$ (Epp 6.3.2)

$$(A \cup B) - C$$

$$= (A \cup B) \cap C^c$$

set difference law

$$= C^c \cap (A \cup B)$$

commutative law for $\cap$

$$= (C^c \cap A) \cup (C^c \cap B)$$

distributive law

$$= (A \cap C^c) \cup (B \cap C^c)$$

commutative law for $\cap$

$$= (A - C) \cup (B - C)$$

set difference law

Worked algebraic proof 2

Prove for all sets A, B: $A - (A \cap B) = A - B$ (Epp 6.3.3)

$$A - (A \cap B)$$

$$= A \cap (A \cap B)^c$$

set difference law

$$= A \cap (A^c \cup B^c)$$

De Morgan's law

$$= (A \cap A^c) \cup (A \cap B^c)$$

distributive law

$$= \emptyset \cup (A \cap B^c)$$

complement law

$$= (A \cap B^c) \cup \emptyset$$

commutative law for $\cup$

$$= A \cap B^c$$

identity law for $\cup$

$$= A - B$$

set difference law

Symmetric difference (enrichment)

Definition

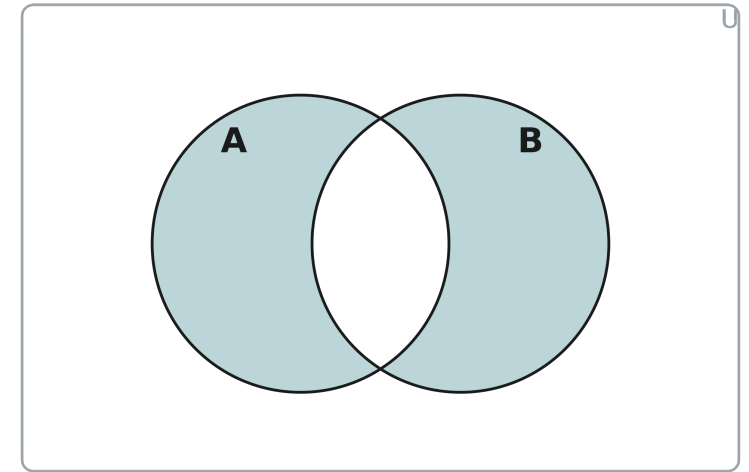
$$A \triangle B = (A - B) \cup (B - A)$$

the elements in exactly one of A, B (not both).

A few properties (exercises 47-50)

$$A \triangle B = B \triangle A \quad A \triangle \emptyset = A$$

$$A \triangle A = \emptyset \quad A \triangle A^c = U$$



A $\triangle$ B shaded

6.4 Boolean Algebra - outline

- 1. One structure behind logic and sets
- 2. The definition of a Boolean algebra (5 axioms)
- 3. The smallest example: $B = \{0, 1\}$
- 4. Properties and the duality principle

(Optional appendix at the end: Russell's paradox and the halting problem.)

One structure behind logic and sets

Compare the laws from Chapter 2 (logic) and Theorem 6.2.2 (sets). They are identical in shape.

Logic	Sets	Boolean
$p \vee q \equiv q \vee p$	$A \cup B = B \cup A$	$a + b = b + a$
$p \wedge q \equiv q \wedge p$	$A \cap B = B \cap A$	$a \cdot b = b \cdot a$
$p \vee c \equiv p$	$A \cup \emptyset = A$	$a + 0 = a$
$p \wedge t \equiv p$	$A \cap U = A$	$a \cdot 1 = a$
$p \vee \sim p \equiv t$	$A \cup A^c = U$	$a + \bar{a} = 1$
$p \wedge \sim p \equiv c$	$A \cap A^c = \emptyset$	$a \cdot \bar{a} = 0$

Correspondence: $\vee \leftrightarrow \cup \leftrightarrow +$ $\wedge \leftrightarrow \cap \leftrightarrow \cdot$ $t \leftrightarrow U \leftrightarrow 1$ $c \leftrightarrow \emptyset \leftrightarrow 0$ $\sim \leftrightarrow ^c \leftrightarrow \bar{}$

Definition of a Boolean algebra (1 of 2)

Setup

A Boolean algebra is a set B with two operations $+$ and $\cdot$ such that for all a, b in B , both $a + b$ and $a \cdot b$ are in B , and these laws hold:

1. Commutative

$$a + b = b + a \quad a \cdot b = b \cdot a$$

2. Associative

$$(a + b) + c = a + (b + c) \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$$

3. Distributive

$$a + (b \cdot c) = (a + b) \cdot (a + c) \quad a \cdot (b + c) = (a \cdot b) + (a \cdot c)$$

Definition of a Boolean algebra (2 of 2)

4. Identity laws

There exist two distinct elements 0 and 1 in B such that for all a :

$$a + 0 = a \qquad a \cdot 1 = a$$

5. Complement laws

For each a in B there is an element $\bar{a}$ (the complement of a) such that:

$$a + \bar{a} = 1 \qquad a \cdot \bar{a} = 0$$

The smallest Boolean algebra: $B = \{0, 1\}$

With these tables, $B = \{0,1\}$ satisfies all five axioms. This is the algebra of a single bit - the basis of digital logic.

+ (acts like OR)

+	0	1
0	0	1
1	1	1

· (acts like AND)

·	0	1
0	0	0
1	0	1

Complement

$$\bar{0} = 1$$

$$\bar{1} = 0$$

0 = false / off

1 = true / on

Notice: $1 + 1 = 1$, not 2. In a Boolean algebra “+” is OR, not ordinary addition.

Theorem 6.4.1: properties of any Boolean algebra

Derivable from the five axioms alone - so they hold for sets, logic, and bits at once.

Uniqueness the complement, 0, and 1 are unique

Double comp. $(\bar{a}) = a$

Idempotent $a + a = a$ $a \cdot a = a$

Universal bound $a + 1 = 1$ $a \cdot 0 = 0$

De Morgan $\overline{a + b} = \bar{a} \cdot \bar{b}$

De Morgan $\overline{a \cdot b} = \bar{a} + \bar{b}$

Absorption $(a + b) \cdot a = a$ $(a \cdot b) + a = a$

Comp. of 0, 1 $\bar{0} = 1$ $\bar{1} = 0$

Worked proof: the double complement law

Prove: $(\bar{a}) = a$ for every a in a Boolean algebra (Epp 6.4.1)

Idea

Complements are unique. So if a satisfies the complement equations for $\bar{a}$, then a must BE the complement of $\bar{a}$, i.e. $(\bar{a}) = a$.

Verify the two complement equations for $\bar{a}$:

$$\bar{a} + a = a + \bar{a} = 1$$

commutative, then complement law

$$\bar{a} \cdot a = a \cdot \bar{a} = 0$$

commutative, then complement law

$\therefore$ by uniqueness of complements, $(\bar{a}) = a$

Worked proof: an idempotent law

Prove: $a + a = a$ for every a in a Boolean algebra (Epp 6.4.2)

Read each step's justification on the right

a

$= a + 0$

0 is the identity for +

$= a + (a \cdot \bar{a})$

complement law for $\cdot$

$= (a + a) \cdot (a + \bar{a})$

distributive law (+ over $\cdot$)

$= (a + a) \cdot 1$

complement law for +

$= a + a$

1 is the identity for $\cdot$

Reading top to bottom: $a = a + a$, hence $a + a = a$.

The duality principle

Principle

Swap $+$ $\leftrightarrow$ $\cdot$ and $0 \leftrightarrow 1$ throughout any Boolean identity. The result (its DUAL) is also a valid identity - automatically, with no new proof.

An identity	Its dual
$a + 0 = a$	$a \cdot 1 = a$
$a + \bar{a} = 1$	$a \cdot \bar{a} = 0$
$a + 1 = 1$	$a \cdot 0 = 0$
$a + (b \cdot c) = (a+b) \cdot (a+c)$	$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

A little history: from al-Khwarizmi to Boole

The word “algebra”

It comes from al-jabr, in the title of al-Khwarizmi's 9th-century book on solving equations. “Algebra” originally meant manipulating symbols by fixed rules.

Boole's leap (1847)

George Boole applied that same symbolic method to LOGIC, in The Mathematical Analysis of Logic. Truth and reasoning became calculation - the foundation of every digital computer.

Recommended exercises (Epp, 4th ed.)

6.3 Find a counterexample	1, 2, 3, 4
6.3 Prove or disprove	5, 6, 7, 9, 10, 13
6.3 Algebraic proofs (cite each step)	30, 31, 32, 34, 35, 38
6.3 Simplify an expression	41, 42, 43
6.3 Symmetric difference (enrichment)	46, 47, 48, 49, 50
6.4 Boolean algebra proofs from axioms	1, 2, 3, 4, 5
6.4 Work in $B = \{0, 1\}$	10, 11

Summary

- To DISPROVE a universal set property, one counterexample is enough.
- A set with n elements has 2^n subsets (proved by induction).
- Algebraic proofs chain the identities of Theorem 6.2.2, citing one law per step.
- Logic, sets, and bits all obey the same laws - they are Boolean algebras.
- A Boolean algebra is a set with $+$, $\cdot$, complement, 0 , 1 , defined by five axioms.
- Duality: swapping $+$ $\leftrightarrow$ $\cdot$ and $0 \leftrightarrow 1$ turns any identity into another valid one.

Optional appendix

Beyond the syllabus: two famous paradoxes

Russell's paradox and the halting problem - both use the same self-reference trick. Enrichment only; not examinable.

Appendix: Russell's paradox

The set that breaks naive set theory

Some sets are not members of themselves (the set of all integers is not an integer). Let:

$S = \{ A \mid A \text{ is a set and } A \notin A \}$. Is $S \in S$?

Both answers contradict

If $S \in S$, then S meets its own defining rule, so $S \notin S$.

If $S \notin S$, then S meets the rule, so $S \in S$.

Either way - contradiction. So “S” cannot be a set.

Appendix: the halting problem

Turing's question (1936)

Could one program `CheckHalt(X, D)` decide, for EVERY program `X` and input `D`, whether `X` halts or loops forever?

Theorem 6.4.2: No such program can exist.

The self-reference trap

Build `Test(X)`: if `CheckHalt(X, X)` says “halts”, loop forever; else halt.

Now run `Test(Test)`: if it halts, `CheckHalt` said “halts”, so it loops; if it loops, it halts.

Contradiction - so `CheckHalt` cannot exist.